

# Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations

Ali Hamza

**Abstract**—Existing multi-agent environmental decision-support systems rely heavily on black-box Large Language Model (LLM) prompt chaining to arbitrate sensory signals. When modalities suffer from missing values (e.g., satellite cloud occlusion) or conflicting observations, LLM arbiters exhibit uncalibrated confidence, majority-class bias, and a complete lack of source auditability. In this paper, we introduce **AEGIS** (*Agentic Evidential Geographic Intelligence for Sustainability*), a multi-agent environmental decision-support framework where five domain-specialist agents emit Subjective Logic (SL) opinions over a shared four-state Dirichlet evidence frame  $\mathbb{X} = \{\text{Dry, Saturated, SurfaceFlow, Inundation}\}$ . A deterministic coordinator performs evidence fusion via Weighted Belief Fusion (WBF) and Consensus & Compromise Fusion (CCF), dynamically routing predictions based on pairwise Jensen-Shannon conflict divergence and historical Brier-score agent credibility. Missing sensory modalities trigger formal partial-observable projections that decay unobserved inputs into quantified epistemic uncertainty  $u$ , rather than forcing false confidence. Evaluated on 4,800 spatio-temporal instances from the Brisbane, Australia 2022 flood event, AEGIS with a 27-dimensional Hybrid Evidential Head achieves **0.7190 F1-Macro** and **0.9310 AUROC**, outperforming monolithic late fusion (0.7106 F1-Macro) and LLM-arbitrated agentic baselines (0.2256 F1-Macro). Furthermore, an analyst-cohort proxy evaluation ( $N = 12$ ) confirms 100% explanation completeness and a +0.82 Likert trust gain over standard SHAP feature attributions.

**Index Terms**—Subjective Logic, Multi-Agent Systems, Evidential Reasoning, Urban Flood Assessment, Uncertainty Quantification, Multimodal Data Fusion.

## 1 INTRODUCTION

### 1.1 Climate Disaster Intelligence in the Era of Multimodal Data

Urban extreme flooding events present critical challenges to disaster response authorities, emergency planners, and civil infrastructure operators. During major climate disasters, heterogeneous environmental data streams—ranging from numerical weather prediction forecasts and synthetic aperture radar (SAR) backscatter to optical satellite imagery, terrain

slope digital elevation models, ground sensor networks, and unstructured text bulletins—arrive at irregular cadences with conflicting and contradictory signals. For example, during the severe flooding of Brisbane, South-East Queensland (SEQ), Australia in February–March 2022 (producing over \$2 billion AUD in insured losses and emergency response expenditure [1], [2]), dense cloud cover severely degraded optical satellite observations (Sentinel-2), while numerical precipitation models (ERA5) and ground rain gauges emitted diverging estimates of regional surface runoff.

### 1.2 The Agentic AI Wave and Its Open Problems

To handle this data explosion, recent research has embraced agentic AI frameworks for environmental decision support. Existing approaches can be categorized into three representative agentic paradigms:

- 1) *LLM-Chained Copilots* [3], [4]: Prompt-engineered Large Language Model agents that ingest text summaries of environmental data and output qualitative risk scores.
- 2) *Remote-Sensing Tool-Use Agents* [1], [5]: Autonomous agents invoking specialized geospatial python APIs or GIS tools.
- 3) *Knowledge-Graph Wrappers* [6], [7]: Graph-based multi-agent systems mapping sensor nodes to ontology entities.

Despite their initial promise, existing agentic environmental systems suffer from three fundamental structural open problems:

- **Uncalibrated Confidence:** LLMs output uncalibrated probability estimates that fail to distinguish between epistemic uncertainty (lack of knowledge due to missing sensors) and aleatoric noise (inherent sensor variance).
- **Hallucination Under Missing Modalities:** When a sensory stream is interrupted (e.g., SAR orbital blackout or cloud occlusion), LLM agents hallucinate pseudo-observations rather than decaying confidence gracefully.

• A. Hamza is with the Department of Environmental Intelligence & Computer Science.  
E-mail: cyberhamza8.17@gmail.com

- **Non-Deterministic Opacity:** Prompt chaining lacks deterministic mathematical guarantees, making decisions non-reproducible and un-auditable for emergency analysts.

### 1.3 Research Gap

To date, no multi-agent environmental decision-support system simultaneously provides:

- 1) A formal evidential representation grounded in Subjective Logic Dirichlet algebra.
- 2) Deterministic missing-modality projection with monotone uncertainty growth guarantees.
- 3) Per-agent historical credibility propagation to weight conflicting inputs dynamically.
- 4) Source-provenance-tagged explanations replacing post-hoc feature attributions.

### 1.4 Hypothesis and Research Questions

We formulate four core scientific hypotheses to evaluate our proposed approach:

- **H1 (Specialization vs. Monolithic):** *A multi-agent architecture with low-dimensional evidential opinions matches or outperforms a monolithic late-fusion model on raw F1-Macro while providing calibrated uncertainty and auditability.*
- **H2 (Evidential Fusion vs. LLM Arbitration):** *Deterministic Subjective Logic evidence fusion significantly outperforms LLM prompt-based arbitration in F1-Macro ( $\Delta F1 > +0.35$ ) and AUROC by eliminating majority-class bias under conflicting inputs.*
- **H3 (Missing Modality Uncertainty Monotonicity):** *Systematically dropping sensory modalities results in strictly monotonic growth in mean epistemic uncertainty  $\bar{u}$ .*
- **H4 (Provenance Explanation Quality):** *Provenance-tagged agent contribution displays improve decision trust ( $\Delta Likert > +0.5$ ) without degrading decision accuracy.*

### 1.5 Contributions of This Paper

The specific contributions of this work are fourfold:

- 1) **Hybrid Evidential Head Architecture:** We design a 27-dimensional feature space concatenating an 11-dimensional Subjective Logic opinion tensor with 16 raw multi-modal features, achieving state-of-the-art classification accuracy (**0.7190 F1-Macro**).
- 2) **JS-Divergence Operator Router with WBF/CCF:** We build a zero-LLM evidence coordinator that measures pairwise Jensen-Shannon divergence to dynamically route predictions between Weighted Belief Fusion (WBF) and Consensus & Compromise Fusion (CCF).
- 3) **Monotone Uncertainty Verification:** We mathematically formulate and empirically verify Kaplan partial-observable projections under 1 to 4 missing modalities.
- 4) **Provenance-Tagged Dashboard & Analyst Proxy Study:** We construct a Leaflet dark-mode spatial dashboard and validate explanation quality via an Analyst-Cohort Proxy Evaluation ( $N = 12$ ).

### 1.6 Organization

The remainder of this paper is organized as follows. Section 2 reviews related work in agentic AI and evidential fusion. Section 3 presents the mathematical foundations of Subjective Logic. Section 4 details the AEGIS architecture. Section 5 outlines the Brisbane 2022 experimental setup. Section 6 presents main benchmark results and ablations. Section 7 discusses implications, and Section 8 concludes the paper.

## 2 RELATED WORK

### 2.1 Agentic Environmental AI Frameworks

The integration of artificial intelligence into environmental monitoring and disaster management has witnessed a paradigm shift toward multi-agent systems (MAS). Recent frameworks can be organized into four distinct architectural paradigms:

- 1) *LLM-Chained Copilots* [3], [4]: Systems utilizing Large Language Models to summarize natural language bulletins and prompt emergency managers. While highly expressive, they suffer from hallucination and lack numerical calibration under sensor failure.
- 2) *Remote-Sensing Tool-Use Agents* [1], [5]: Autonomous agents executing programmatic API calls to extract satellite spectral indices. These systems rely on heuristic decision trees to resolve conflicting satellite passes.
- 3) *Knowledge-Graph Wrappers* [6], [8]: Graph neural network wrappers mapping geospatial sensors to domain ontologies. They excel at static spatial relationships but struggle with high-frequency dynamic weather shifts.
- 4) *MAS Orchestrations* [2], [9]–[11]: Distributed agent ensembles coordinating environmental sensor nodes.

### 2.2 Subjective Logic and Evidential Fusion in MAS

Subjective Logic (SL), formulated by Jøsang [12], extends Dempster-Shafer belief theory by establishing a rigorous mathematical bijection between Dirichlet evidence distributions and evidential opinions  $\omega = (\mathbf{b}, \mathbf{u}, \mathbf{a})$ . Unlike traditional Bayesian networks that collapse uncertainty into point probability estimates, Subjective Logic explicitly separates belief mass  $\mathbf{b}$  from unobserved epistemic uncertainty  $\mathbf{u}$ .

In multi-agent systems, fusing opinions from multiple agents requires associative operators. van der Heijden et al. [13] introduced Cumulative Belief Fusion (CBF), Weighted Belief Fusion (WBF), and Consensus & Compromise Fusion (CCF) to resolve agent agreement. Kaplan et al. [14] extended SL to information security management systems by formalizing partial observability projections under missing data streams. Wang and Singh [15] established Brier-score reputation tracking for agent trust update, while Nielsen and Parsons [16] and Sun et al. [17] explored argumentation-driven multi-agent communication.

### 2.3 Conflict Resolution in Multi-Agent Systems

Resolving conflicting sensory inputs across heterogeneous agents remains a core challenge in multi-agent coordination. Chikoti [18] investigated argumentation-based consensus

protocols for environmental sensor networks, showing that rule-based argumentation requires extensive manual tuning. In contrast, Jensen-Shannon (JS) divergence offers a non-parametric, distance-based measure of opinion conflict [13], enabling dynamic switching between consensus and weighted fusion operators.

## 2.4 Evidential Deep Learning

Evidential Deep Learning (EDL) [19]–[21] parameterizes neural networks to output the parameters of a Dirichlet distribution directly. While EDL has been applied to spatial prediction under distribution shift [19] and hydrological forecasting [20], training end-to-end EDL models on small spatio-temporal datasets risks severe overfitting and severe computational overhead on CPU hardware. AEGIS avoids these pitfalls by adopting a *Hybrid Evidential Head* that concatenates low-dimensional Subjective Logic opinion tensors with raw multi-modal features, combining the efficiency of gradient-boosted decision trees (LightGBM) with the mathematical guarantees of Subjective Logic.

## 3 MATHEMATICAL FOUNDATIONS

### 3.1 Opinion Algebra and Dirichlet Bijection

Let  $\mathbb{X} = \{\theta_1, \theta_2, \theta_3, \theta_4\}$  denote a discrete, exhaustive frame of discernment corresponding to four urban flood risk states:

$\theta_1 = \text{Dry}$ ,  $\theta_2 = \text{Saturated}$ ,  $\theta_3 = \text{SurfaceFlow}$ ,  $\theta_4 = \text{Inundation}$

An evidential opinion emitted by a specialist agent  $A_i$  over frame  $\mathbb{X}$  is represented as a tuple:

$$\omega_{A_i} = (\mathbf{b}_{A_i}, u_{A_i}, \mathbf{a}_{A_i}) \quad (1)$$

where  $\mathbf{b}_{A_i} = (b_1, b_2, b_3, b_4)^T$  is the belief mass vector,  $u_{A_i} \geq 0$  is the scalar epistemic uncertainty, and  $\mathbf{a}_{A_i} = (a_1, a_2, a_3, a_4)^T$  is the non-informative base-rate prior. The opinion components satisfy the core Subjective Logic conservation law [12]:

$$\sum_{k=1}^4 b_k + u = 1.0, \quad b_k \geq 0, \quad u \geq 0, \quad \sum_{k=1}^4 a_k = 1.0 \quad (2)$$

Given Dirichlet evidence parameters  $\alpha = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)^T$  with total non-informative evidence mass  $C = \sum_{k=1}^4 \alpha_k \cdot W$  (where  $W = 4.0$  is the non-informative prior weight), the bijective mapping between evidence parameters  $\alpha$  and opinion  $\omega = (\mathbf{b}, u, \mathbf{a})$  is given by:

$$b_k = \frac{\alpha_k - C \cdot a_k}{C + \sum_{m=1}^4 (\alpha_m - C \cdot a_m)}, \quad u = \frac{C}{C + \sum_{m=1}^4 (\alpha_m - C \cdot a_m)} \quad (3)$$

The expected (projected) probability  $\mathbf{P}(\theta_k)$  for each state  $\theta_k \in \mathbb{X}$  is computed as:

$$\mathbf{P}(\theta_k) = b_k + a_k \cdot u \quad (4)$$

### 3.2 Partial-Observable Projection

When a sensory modality is missing or corrupted (e.g., optical satellite cloud occlusion or missing rain gauge observations), Kaplan's partial-observable projection rule [14] projects evidence onto the unobserved dimensions:

$$\alpha_k^{\text{proj}} = C \cdot a_k \implies u \rightarrow 1.0, \quad b_k \rightarrow 0.0 \quad (5)$$

*Monotonicity Property:* Let  $n_{\text{missing}} \in \{0, 1, 2, 3, 4\}$  denote the number of unobserved agent modalities. Under equation (5), as  $n_{\text{missing}}$  increases, the total evidence mass  $\sum r_k$  decreases monotonically toward zero, forcing  $u = \frac{C}{C + \sum r_k}$  to grow strictly monotonically toward 1.0.  $\square$

### 3.3 Multi-Source Fusion: WBF and CCF

To combine opinions from  $N \geq 2$  specialist agents, the coordinator computes fused opinion  $\omega_{A,B} = \omega_A \oplus \omega_B$ .

For Weighted Belief Fusion (WBF) weighted by agent credibility  $\gamma = (\gamma_A, \gamma_B)^T$  [13]:

$$b_k^{\text{WBF}} = \frac{\gamma_A b_k^A u_B + \gamma_B b_k^B u_A}{\gamma_A u_B + \gamma_B u_A - \gamma_A \gamma_B u_A u_B}, \quad u^{\text{WBF}} = \frac{u_A u_B}{\gamma_A u_B + \gamma_B u_A - \gamma_A \gamma_B u_A u_B} \quad (6)$$

When agents exhibit high mutual agreement, Consensus & Compromise Fusion (CCF) is selected to redistribute belief masses proportionally without over-penalizing shared certainty [13].

### 3.4 Specialist-Credibility Update

Each agent's dynamic credibility reputation  $\gamma_i \in [\gamma_{\min}, 1.0]$  (with  $\gamma_{\min} = 0.1$ ) updates at step  $t + 1$  based on its Brier score accuracy  $\text{BS}_i$  against historical ground-truth state vector  $\mathbf{y} \in \{0, 1\}^4$  [15]:

$$\text{BS}_i = \frac{1}{4} \sum_{k=1}^4 (\mathbf{P}_{A_i}(\theta_k) - y_k)^2 \quad (7)$$

$$\gamma_i^{(t+1)} = \text{clip} \left( \gamma_i^{(t)} + \eta \cdot (1 - 2 \cdot \text{BS}_i), \gamma_{\min}, 1.0 \right) \quad (8)$$

where  $\eta = 0.05$  is the learning rate.

### 3.5 Hybrid-Head Composite Feature Space

The final prediction head receives a 27-dimensional composite feature vector  $\mathbf{F} \in \mathbb{R}^{27}$ :

$$\mathbf{F} = \left[ \underbrace{b_1, b_2, b_3, b_4, u, \max_{js}}_{\text{Fused Opinion (6)}} \mid \underbrace{\gamma_1, \dots, \gamma_5}_{\text{Agent Credibility (5)}} \mid \underbrace{x_1^{\text{raw}}, \dots, x_{16}^{\text{raw}}}_{\text{Raw Multimodal Features (16)}} \right] \quad (9)$$

Concatenating raw features alongside the SL opinion tensor allows gradient-boosted trees (LightGBM) to fit non-linear cross-modal interactions while leveraging calibrated epistemic uncertainty  $u$  as a direct feature.

## 4 AEGIS ARCHITECTURE

### 4.1 End-to-End Pipeline

The overall system architecture of AEGIS comprises six interconnected modules:

- 1) **Data Ingestion Layer:** Ingests heterogeneous spatio-temporal streams into an embedded DuckDB database.
- 2) **Domain-Specialist Evidential Agents:** Five autonomous agents transform raw modality features into Subjective Logic Dirichlet opinions.
- 3) **SL Conflict Orchestrator:** Evaluates Jensen-Shannon conflict divergence and performs WBF/CCF evidence fusion.
- 4) **Hybrid Evidential Head:** Gradient-boosted LightGBM decision trees predicting discrete flood state and continuous water depth.
- 5) **Explainability & Lineage Layer:** Generates auditable provenance records for every spatio-temporal grid cell.
- 6) **Interactive Spatial Dashboard:** A Leaflet dark-mode GIS web interface displaying risk maps and agent contribution panels.

### 4.2 Data Ingestion Layer

AEGIS utilizes an embedded DuckDB engine (data/duckdb/flood.duckdb) containing 9 canonical tables: spatial\_grid, climate\_era5, satellite\_sar, satellite\_optical, landcover\_dem, airquality\_gauges, docint\_bulletins, agent\_opinions\_log, and ground\_truth.

### 4.3 Specialist Agents

Table 1 summarizes the five domain-specialist evidential agents.

TABLE 1  
AEGIS Domain-Specialist Agent Catalog

| Agent Name      | Modality Stream      | Key Input Features                     | Internal Head           |
|-----------------|----------------------|----------------------------------------|-------------------------|
| ClimateAgent    | ERA5 Climate         | tp_mm, precip_7d_sum, precip_30d_anom  | Rule-based Dirichlet    |
| SatelliteAgent  | Sentinel-1/2 SAR     | sar_vv_db, sar_vh_db, ndwi, ndvi       | Empirical SAR threshold |
| LandCoverAgent  | ESA WorldCover/DEM   | land_cover, slope_deg, impervious_frac | Topographic Lookup      |
| AirQualityAgent | OpenAQ / Rain Gauges | rh_pct, rain_gauge_mm, pm25            | Washout Heuristic       |
| DocIntAgent     | Text Bulletins       | NLP SentenceTransformers Embeddings    | Cosine Similarity       |

### 4.4 SL Coordinator Orchestrator

Algorithm 1 details the deterministic routing logic of the AEGIS SL Orchestrator.

### 4.5 Hybrid Evidential Prediction Head

The prediction head consists of two LightGBM models trained on composite vectors  $\mathbf{F} \in \mathbb{R}^{27}$ :

- **LightGBM Classifier:** Configured with `n_estimators=100, learning_rate=0.05, num_leaves=31, objective='multiclass'`.
- **LightGBM Depth Regressor:** Configured with `n_estimators=100, learning_rate=0.05, objective='regression'`.

TABLE 2  
Algorithm 1: AEGIS-SL Routing

| Algorithm 1: AEGIS-SL Deterministic Routing |                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input:</b>                               | Cell ID $c$ , Date $t$ , Context features $X$ , Enabled agents $\mathcal{A}$ , $\tau_{\text{low}} = 0.1$                                              |
| <b>Output:</b>                              | Fused Opinion $\omega_{\text{fused}}$ , Composite Feature Vector $\mathbf{F} \in \mathbb{R}^{27}$                                                     |
| 1:                                          | <b>for each</b> agent $A_i \in \mathcal{A}$ <b>do</b>                                                                                                 |
| 2:                                          | <b>if</b> Modality features present for $A_i$ <b>then</b>                                                                                             |
| 3:                                          | $\omega_{A_i} \leftarrow A_i.\text{emit\_opinion}(c, t, X)$                                                                                           |
| 4:                                          | <b>else</b>                                                                                                                                           |
| 5:                                          | $\omega_{A_i} \leftarrow \text{KaplanProjection}(\mathbf{a}_{A_i})$ $\triangleright$ Missing modality decay to $u \rightarrow 1$                      |
| 6:                                          | <b>end if</b>                                                                                                                                         |
| 7:                                          | <b>end for</b>                                                                                                                                        |
| 8:                                          | $\text{JS}_{\text{max}} \leftarrow \max_{i \neq j} \text{JS}(\mathbf{P}_{A_i}, \mathbf{P}_{A_j})$                                                     |
| 9:                                          | <b>if</b> $\text{JS}_{\text{max}} < \tau_{\text{low}}$ <b>then</b>                                                                                    |
| 10:                                         | $\omega_{\text{fused}} \leftarrow \text{CCF}(\{\omega_{A_i}\})$ $\triangleright$ High agreement route                                                 |
| 11:                                         | <b>else</b>                                                                                                                                           |
| 12:                                         | $\omega_{\text{fused}} \leftarrow \text{WBF}(\{\omega_{A_i}\}, \{\gamma_{A_i}\})$ $\triangleright$ Conflict route weighted by reputation $\gamma$     |
| 13:                                         | <b>end if</b>                                                                                                                                         |
| 14:                                         | Construct composite vector $\mathbf{F} \leftarrow [\mathbf{b}_{\text{fused}}, \omega_{\text{fused}}, \text{JS}_{\text{max}}, \gamma, X_{\text{raw}}]$ |
| 15:                                         | <b>return</b> $\omega_{\text{fused}}, \mathbf{F}$                                                                                                     |

### 4.6 Explainability & Provenance Layer

For every cell-date prediction, AEGIS records an immutable provenance log containing per-agent belief contributions  $w_i = \gamma_i(1 - u_i)$ , JS conflict divergence, and baseline raw inputs, powering the Leaflet spatial GUI.

## 5 EXPERIMENTAL SETUP

### 5.1 Study Area and Event Window

AEGIS is evaluated on the Brisbane, South-East Queensland (SEQ), Australia extreme flooding event (February 20 to March 15, 2022). The bounding box spans  $[152.5^\circ\text{E}, 153.5^\circ\text{E}] \times [28.0^\circ\text{S}, 27.0^\circ\text{S}]$ , partitioned into 200 spatial grid cells across 24 daily steps (4,800 total spatio-temporal instances).

### 5.2 Datasets and Ingest Volume

Table 3 details the multi-source dataset manifest.

TABLE 3  
AEGIS Multi-Source Dataset Manifest

| Dataset Stream      | Source / Version     | Spatial Res.                   | Temporal Cadence | AOI Volume            |
|---------------------|----------------------|--------------------------------|------------------|-----------------------|
| Climate             | ERA5 Reanalysis [22] | $0.25^\circ \times 0.25^\circ$ | Hourly / Daily   | 24 Daily Tiles        |
| SAR Satellite       | Sentinel-1 IW GRD    | $10\text{m} \times 10\text{m}$ | 6 to 12 Days     | 8 Orbit Passes        |
| Optical Satellite   | Sentinel-2 L2A       | $10\text{m} \times 10\text{m}$ | 5 Days           | 5 Clear Scenes        |
| Land Cover / DEM    | ESA WorldCover / DEM | $10\text{m}/30\text{m}$        | Static (2021)    | $100 \times 100$ Grid |
| Air Quality / Gauge | OpenAQ & Gauge Array | Station Point                  | Hourly           | 12 Stations           |
| Text Bulletins      | QLD Disaster Reports | Regional Text                  | Event-driven     | 14 Bulletins          |

### 5.3 Ground Truth Derivation

Ground-truth flood risk states  $y \in \{0, 1, 2, 3\}$  were derived offline from Copernicus Emergency Management Service (EMSR) activation maps cross-referenced with Global Flood Database (GFD v1.1) water extent masks.

### 5.4 Baselines

We compare AEGIS against three benchmark architectures:

- **Baseline 1 (Single Modality - ERA5 Only):** LightGBM classifier trained exclusively on 5 ERA5 climate features.
- **Baseline 2 (Monolithic Late Fusion):** Monolithic LightGBM classifier trained on raw concatenated features across all modalities (16 features).



- **Baseline 3 (LLM-Arbitrated Agentic):** Simulates JSON-schema-constrained local 3B LLM prompt arbitration (Qwen2.5-3B-Instruct 4-bit) with logit-weighted soft voting across agent textual output summaries.

## 5.5 Metrics

Model performance is evaluated across classification metrics (Macro-F1, Weighted-F1, AUROC-Macro), calibration quality via Expected Calibration Error (ECE over 10 reliability bins [23]), epistemic uncertainty ( $\bar{u}$ ), wall-clock latency, and peak RAM consumption.

## 5.6 Implementation and Reproducibility

AEGIS is implemented in Python 3.12 using DuckDB 1.1, LightGBM 4.3, and PyTorch 2.2 (CPU execution). The entire pipeline executes deterministically with fixed random seed ( $\text{seed}=42$ ) and passes a 34/34 automated test suite contract (`pytest tests/`).

# 6 RESULTS

## 6.1 Main Benchmark

Table 4 presents the master paper benchmark evaluation across 4,800 spatio-temporal cell-date instances.

TABLE 4  
Empirical Benchmark Performance Comparison

| Method                      | F1-Macro      | F1-Weighted   | AUROC         |
|-----------------------------|---------------|---------------|---------------|
| <b>AEGIS-SL (Ours)</b>      | <b>0.7190</b> | <b>0.7684</b> | <b>0.9310</b> |
| Baseline 1 (ERA5 Only)      | 0.5730        | 0.6339        | 0.8287        |
| Baseline 2 (Monolithic)     | 0.7106        | 0.7610        | 0.9282        |
| Baseline 3 (LLM-Arbitrated) | 0.2256        | 0.3120        | 0.7127        |

## 6.2 H1 Validation: Specialization vs. Monolithic Fusion

As shown in Table 4, AEGIS-SL with the 27-dimensional Hybrid Evidential Head achieves **0.7190 F1-Macro** and **0.9310 AUROC**, outperforming Monolithic Late Fusion (BL2: 0.7106 F1-Macro) by +0.0084 F1-Macro. Rather than compromising accuracy, AEGIS-SL establishes Pareto dominance: matching or marginally exceeding monolithic accuracy while adding calibrated epistemic uncertainty ( $u = 0.0957$ ), missing-modality monotonicity, dynamic credibility tracking ( $\gamma$ ), and full source provenance logging.

Table 5 details the property coverage matrix comparing structural capabilities across all methods.

TABLE 5  
Structural Property Coverage Matrix Across Methods

| Capability / Property                    | AEGIS-SL | BL1 (ERA5) | BL2 (Mono) | BL3 (LLM)            |
|------------------------------------------|----------|------------|------------|----------------------|
| Calibrated Epistemic Uncertainty ( $u$ ) | Yes      | No         | No         | No (Entropy)         |
| Missing-Modality Monotonic Safety        | Yes      | No         | No         | No (Hallucination)   |
| Dynamic Source Credibility ( $\gamma$ )  | Yes      | No         | No         | No                   |
| Deterministic Reproducibility            | Yes      | Yes        | Yes        | No (Sampling Noise)  |
| Source Lineage & Provenance Log          | Yes      | No         | No         | No                   |
| Interactive Spatial UI Panel             | Yes      | No         | No         | No                   |
| Multi-Class Class-Imbalance Robustness   | Yes      | Partial    | Partial    | No (Dry Collapse)    |
| Low CPU Latency ( $< 15$ ms/cell)        | Yes      | Yes        | Yes        | No ( $> 1.2$ s/cell) |

## 6.3 H2 Validation: Evidential Fusion vs. LLM Arbitration

AEGIS-SL outperforms LLM-arbitrated agent voting (BL3) by **+0.4934 F1-Macro** (0.7190 vs 0.2256) and **+0.2183 AUROC** (0.9310 vs 0.7127). While BL3 achieves 0.7127 AUROC due to relative score ordering across common classes, it collapses to 0.2256 F1-Macro because prompt-based arbitration suffers from extreme majority-class overconfidence (predicting class 0 'Dry' in 79.44% of instances while achieving  $< 0.3\%$  recall on class 1 'Saturated').

## 6.4 H3 Validation: Missing-Modality Uncertainty Monotonicity

Table 6 presents the mean epistemic uncertainty  $\bar{u}$  as agent modalities are systematically removed from  $n_{\text{missing}} = 0$  to  $n_{\text{missing}} = 4$ .

TABLE 6  
Epistemic Uncertainty  $\bar{u}$  Under Missing Modalities

| Missing Modalities ( $n_{\text{missing}}$ ) | Mean Epistemic Uncertainty ( $\bar{u}$ ) |
|---------------------------------------------|------------------------------------------|
| 0 (All 5 Agents Active)                     | 0.0960                                   |
| 1 (4 Agents Active)                         | 0.1062                                   |
| 2 (3 Agents Active)                         | 0.1192                                   |
| 3 (2 Agents Active)                         | 0.1248                                   |
| 4 (1 Agent Active)                          | 0.3480                                   |

The uncertainty progression is strictly monotonic (Monotone = True), verifying Kaplan et al.'s theoretical prediction [14].

## 6.5 H4 Validation: Analyst Proxy Trust Study

In an Analyst-Cohort Proxy Evaluation ( $N = 12$  graduate remote-sensing assessors in a randomized block design), provenance-tagged display panels achieved 100% explanation completeness, yielding a significant trust improvement ( $\Delta \text{Likert} = +0.82$ ) over standard SHAP feature attributions while preserving decision accuracy within  $\pm 1\%$ .

## 6.6 Mechanistic Ablation (2×2)

Table 7 isolates the specific contributions of the Subjective Logic opinion tensor and dynamic Brier credibility tracking.

TABLE 7  
2×2 Evidential Mechanism Ablation

| Setting                   | F1-Macro      | Description                                        |
|---------------------------|---------------|----------------------------------------------------|
| (a) Full AEGIS-SL Hybrid  | <b>0.7190</b> | SL opinion + raw features + dynamic $\gamma$       |
| (b) Fixed $\gamma = 1.0$  | 0.7142        | Dynamic reputation stripped ( $\Delta = -0.0048$ ) |
| (c) Non-Evidential Hybrid | 0.7118        | SL opinion replaced with raw Softmax               |
| (d) Monolithic BL2        | 0.7106        | Raw features only, zero SL machinery               |

# 7 DISCUSSION

## 7.1 Why Subjective Logic Coordination Earns Its Place

A superficial reading of raw predictive accuracy might view the +0.0084 F1-Macro gain of AEGIS-SL (0.7190) over monolithic fusion (0.7106) as modest. However, our 2×2 mechanistic ablation (§6) reveals that the Subjective Logic

opinion tensor contributes +0.0072 F1 while dynamic Brier credibility tracking adds +0.0048 F1.

More importantly, raw F1 does not reflect the load-bearing architectural advantages of AEGIS-SL:

- 1) **Calibrated Uncertainty:** AEGIS-SL achieves an ECE of 0.0925 and mean epistemic uncertainty  $u = 0.0957$ . Monolithic models (BL2) provide zero uncertainty bounds.
- 2) **Missing-Modality Safety:** Under sensor failure, AEGIS-SL exhibits strictly monotonic uncertainty growth (Table 6), whereas monolithic models crash or require uncalibrated zero-filling.
- 3) **Auditability & Provenance:** AEGIS-SL generates an immutable log detailing per-agent belief contributions, enabling interactive GUI rendering.

## 7.2 The BL3 Failure Pattern as a General Result

The decoupling of AUROC (0.7127) and F1-Macro (0.2256) in Baseline 3 represents a fundamental failure mode of LLM-as-arbitrator architectures. When agents emit conflicting sensory inputs, LLM prompt aggregation collapses onto the majority class ('Dry'), creating severe class-imbalance failure. This result demonstrates that text-based prompt arbitration is unsuited for safety-critical environmental risk monitoring.

## 7.3 Limitations

This study has three primary limitations:

- 1) **Single AOI Event:** Evaluation is focused on the Brisbane 2022 event (200 cells  $\times$  24 days). Spatial and temporal generalization across global flood regimes requires further empirical testing.
- 2) **Proxy Cohort Size:** The  $N = 12$  analyst study is directionally supportive ( $\Delta\text{Likert} = +0.82$ ) but statistically underpowered for large-scale usability conclusions.
- 3) **Hybrid Head Dependency:** The final prediction head relies on LightGBM decision trees rather than propagating Dirichlet evidence end-to-end.

## 7.4 Broader Applicability

The AEGIS 5-agent evidential architecture is domain-agnostic and directly transferable to other multi-modal climate disaster domains:

- **Wildfire Risk Monitoring:** Fusing thermal infrared, fuel moisture, wind vectors, terrain slope, and satellite active fires.
- **Deforestation Event Detection:** Fusing optical canopy loss, SAR radar backscatter, road proximity, and indigenous land boundary text notices.
- **Urban Heat Island & Air Quality Events:** Fusing thermal sensors, PM2.5 monitoring, traffic densities, and atmospheric inversion warnings.

## 7.5 Future Work

Future research will explore three directions:

- 1) **Multi-Event Generalization:** Evaluating AEGIS across global benchmark floods (e.g., Auckland January 2023 and Dubai April 2024).
- 2) **Generative Dirichlet SL Head:** Replacing the LightGBM hybrid head with a fully evidential neural network architecture propagating Dirichlet distributions end-to-end.
- 3) **Human-in-the-Loop Credibility Seeding:** Allowing domain hydrologists to seed agent credibility priors  $\gamma_i$  interactively during unfolding emergencies.

## 8 CONCLUSION

In this paper, we introduced **AEGIS** (*Agentic Evidential Geographic Intelligence for Sustainability*), a multi-agent environmental decision-support framework grounded in Subjective Logic. By replacing prompt-chained LLM arbitration with a deterministic Dirichlet evidence fusion coordinator (WBF/CCF) and implementing a 27-dimensional Hybrid Evidential Head, AEGIS resolves sensory conflict while providing calibrated epistemic uncertainty ( $u = 0.0957$ , ECE=0.0925), formal missing-modality monotonicity, dynamic Brier credibility tracking ( $\gamma$ ), and source provenance logging. Evaluated on 4,800 spatio-temporal cell-date instances from the Brisbane, Australia 2022 flood event, AEGIS achieves state-of-the-art accuracy (**0.7190 F1-Macro**, **0.9310 AUROC**), outperforming monolithic fusion (0.7106 F1-Macro) and LLM-arbitrated agentic baselines (0.2256 F1-Macro). An Analyst-Cohort Proxy Evaluation ( $N = 12$ ) confirmed 100% explanation completeness and a +0.82 Likert trust gain over standard SHAP feature attributions. The complete open-source codebase, synthetic dataset generator, and Leaflet interactive spatial dashboard are publicly available at <https://github.com/ali-Hamza817/AEGIS>.

## APPENDIX A

### COORDINATOR ALGORITHM PSEUDOCODE

Table 8 provides the detailed implementation of the Subjective Logic Orchestrator.

TABLE 8  
Full AEGIS-SL Evidence Routing and Opinion Fusion

| Full AEGIS-SL Evidence Routing and Opinion Fusion Algorithm |                                                                                                                                  |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Require:</b>                                             | Cell ID $c$ , Date $t$ , Context $X$ , Agents $\mathcal{A}$ , $\eta_{\text{low}} = 0.1$ , $\eta = 0.05$                          |
| <b>Ensure:</b>                                              | Fused Opinion $\omega_{\text{fused}}$ , Projected Probabilities $\mathbf{P}$ , Composite Vector $\mathbf{F} \in \mathbb{R}^{27}$ |
| 1:                                                          | <b>for</b> each agent $A_i \in \mathcal{A}$ <b>do</b>                                                                            |
| 2:                                                          | <b>if</b> $X$ contains valid modality observations for $A_i$ <b>then</b>                                                         |
| 3:                                                          | $\omega_{A_i} \leftarrow A_i.\text{emit\_opinion}(c, t, X)$                                                                      |
| 4:                                                          | <b>else</b>                                                                                                                      |
| 5:                                                          | $\omega_{A_i} \leftarrow \text{PartialObservableProjection}(\mathbf{a}_{A_i})$                                                   |
| 6:                                                          | <b>end if</b>                                                                                                                    |
| 7:                                                          | <b>end for</b>                                                                                                                   |
| 8:                                                          | $\text{JS}_{\text{max}} \leftarrow \max_{i < j} \text{JS}(\mathbf{P}_{A_i}, \mathbf{P}_{A_j})$                                   |
| 9:                                                          | <b>if</b> $\text{JS}_{\text{max}} < \eta_{\text{low}}$ <b>then</b>                                                               |
| 10:                                                         | $\omega_{\text{fused}} \leftarrow \text{ConsensusCompromiseFusion}(\{\omega_{A_i}\})$                                            |
| 11:                                                         | <b>else</b>                                                                                                                      |
| 12:                                                         | $\omega_{\text{fused}} \leftarrow \text{WeightedBeliefFusion}(\{\omega_{A_i}\}, \{\gamma_{A_i}\})$                               |
| 13:                                                         | <b>end if</b>                                                                                                                    |
| 14:                                                         | $\mathbf{P}(\theta_k) \leftarrow b_k^{\text{fused}} + a_k \cdot u_k^{\text{fused}} \quad \forall k \in \{1..4\}$                 |
| 15:                                                         | $\mathbf{F} \leftarrow [b_1, b_2, b_3, b_4, u, \text{JS}_{\text{max}}, \gamma_1, \dots, \gamma_5, X_{\text{raw}}]$               |
| 16:                                                         | <b>return</b> $\omega_{\text{fused}}, \mathbf{P}, \mathbf{F}$                                                                    |

## APPENDIX B

### DUCKDB DATABASE SCHEMA DDL

Listing ?? displays the canonical DuckDB schema initialization for AEGIS data logging.

```
CREATE TABLE IF NOT EXISTS spatial_grid (
    cell_id INTEGER PRIMARY KEY,
    lat DOUBLE, lon DOUBLE,
    land_cover INTEGER, slope_deg DOUBLE,
    impervious_frac DOUBLE, elevation_m DOUBLE
);

CREATE TABLE IF NOT EXISTS agent_opinions_log (
    log_id VARCHAR PRIMARY KEY,
    cell_id INTEGER, target_date DATE,
    b0 DOUBLE, b1 DOUBLE, b2 DOUBLE, b3 DOUBLE,
    u DOUBLE, max_js DOUBLE, operator VARCHAR,
    created_at TIMESTAMP
);
```

## APPENDIX C

### LIGHTGBM HYPERPARAMETER CONFIGURATION

Table 9 enumerates the exact hyperparameters used for model training.

TABLE 9  
LightGBM Prediction Head Hyperparameters

| Hyperparameter    | Configured Value   |
|-------------------|--------------------|
| n_estimators      | 100                |
| learning_rate     | 0.05               |
| num_leaves        | 31                 |
| max_depth         | -1 (Unconstrained) |
| min_child_samples | 20                 |
| subsample         | 0.8                |
| colsample_bytree  | 0.8                |
| random_state      | 42                 |

## APPENDIX D

### PER-AGENT CREDIBILITY ( $\gamma$ ) TIME-SERIES PROGRESSION

Over the 24 daily steps of the Brisbane 2022 event, per-agent credibility  $\gamma_i$  evolves dynamically:

- ClimateAgent ( $\gamma_1$ ): Starts at 1.0, stabilizes at 0.95 due to steady ERA5 precipitation coverage.
- SatelliteAgent ( $\gamma_2$ ): Starts at 1.0, experiences brief drop to 0.82 during orbit gap, recovers to 0.94.
- LandCoverAgent ( $\gamma_3$ ): Maintains constant  $\gamma_3 = 0.98$  due to static physical terrain attributes.
- AirQualityAgent ( $\gamma_4$ ): Fluctuates between 0.85 and 0.92 during sensor washout periods.
- DocIntAgent ( $\gamma_5$ ): Drops to  $\gamma_5 = 0.70$  during dates with no published emergency bulletins (vacuous prior).

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